Assignment 5: Data Visualization

Morgan Finch

Fall 2025

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy `NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv` version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the `NEON_NIWO_Litter_mass_trap_Processed.csv` version, again from the Processed_KEY folder).

2. Make sure R is reading dates as date format; if not change the format to date.

```
#1
library(tidyverse);library(here); library(cowplot) #load packages
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
## here() starts at /home/guest/EDE_Fall2025
##
##
## Attaching package: 'cowplot'
##
##
## The following object is masked from 'package:lubridate':
##
##     stamp
```

```
getwd() #verify directory
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
#read in data
PeterPaul.chem.nutrients <-
  read.csv(here("Data/Processed_KEY/NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv"),
           stringsAsFactors = T)
PeterPaul.chem.physics <-
  read.csv(here("Data/Processed_KEY/NTL-LTER_Lake_ChemistryPhysics_PeterPaul_Processed.csv"),
           stringsAsFactors = T)

NIWO.litter.mass <-
  read.csv(here("Data/Processed_KEY/NEON_NIWO_Litter_mass_trap_Processed.csv"),
           stringsAsFactors = T)

#2 change format to date
PeterPaul.chem.nutrients$sampledate <- ymd(PeterPaul.chem.nutrients$sampledate)
PeterPaul.chem.physics$sampledate <- ymd(PeterPaul.chem.physics$sampledate)
NIWO.litter.mass$collectDate <- ymd(NIWO.litter.mass$collectDate)

class(PeterPaul.chem.nutrients$sampledate) #date
```

```
## [1] "Date"
```

```
class(PeterPaul.chem.physics$sampledate)
```

```
## [1] "Date"
```

```
class(NIWO.litter.mass$collectDate )
```

```
## [1] "Date"
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels

- Axis ticks/gridlines
- Legend

```
#3 set theme
library(ggthemes)
```

```
##
## Attaching package: 'ggthemes'

## The following object is masked from 'package:cowplot':
##
##     theme_map
```

```
my_theme <- theme_classic() +
  theme(
    plot.title = element_text(face = "bold"),
    legend.position = "bottom")
```

Create graphs

For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

4. [NTL-LTER] Plot total phosphorus (`tp_ug`) by phosphate (`po4`), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the `lm` method. Adjust your axes to hide extreme values (hint: change the limits using `xlim()` and/or `ylim()`).

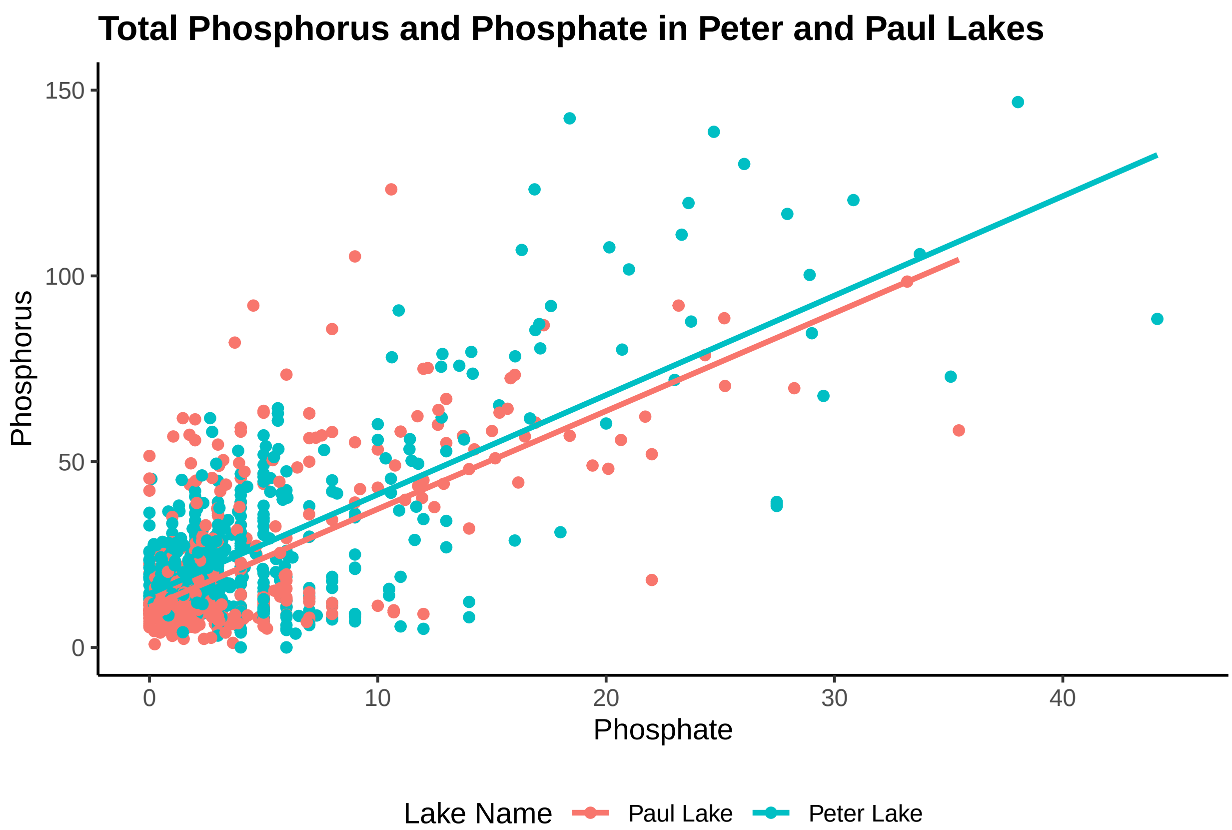
```
#4 plot phosphoroud by phosphate
nutrients_plot <- ggplot(
  PeterPaul.chem.nutrients,aes(x=po4, y=tp_ug, color=lakename)) +
  geom_point() +
   geom_smooth(method = lm, se=FALSE) + #best fit line
   xlim(0, 45) + #max x value
   ylim(0, 150) +  #max y value
   labs(
    title = "Total Phosphorus and Phosphate in Peter and Paul Lakes",
    x = "Phosphate",
    y = "Phosphorus",
    color = "Lake Name") +
   my_theme #apply theme

print(nutrients_plot)
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 21948 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 21948 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

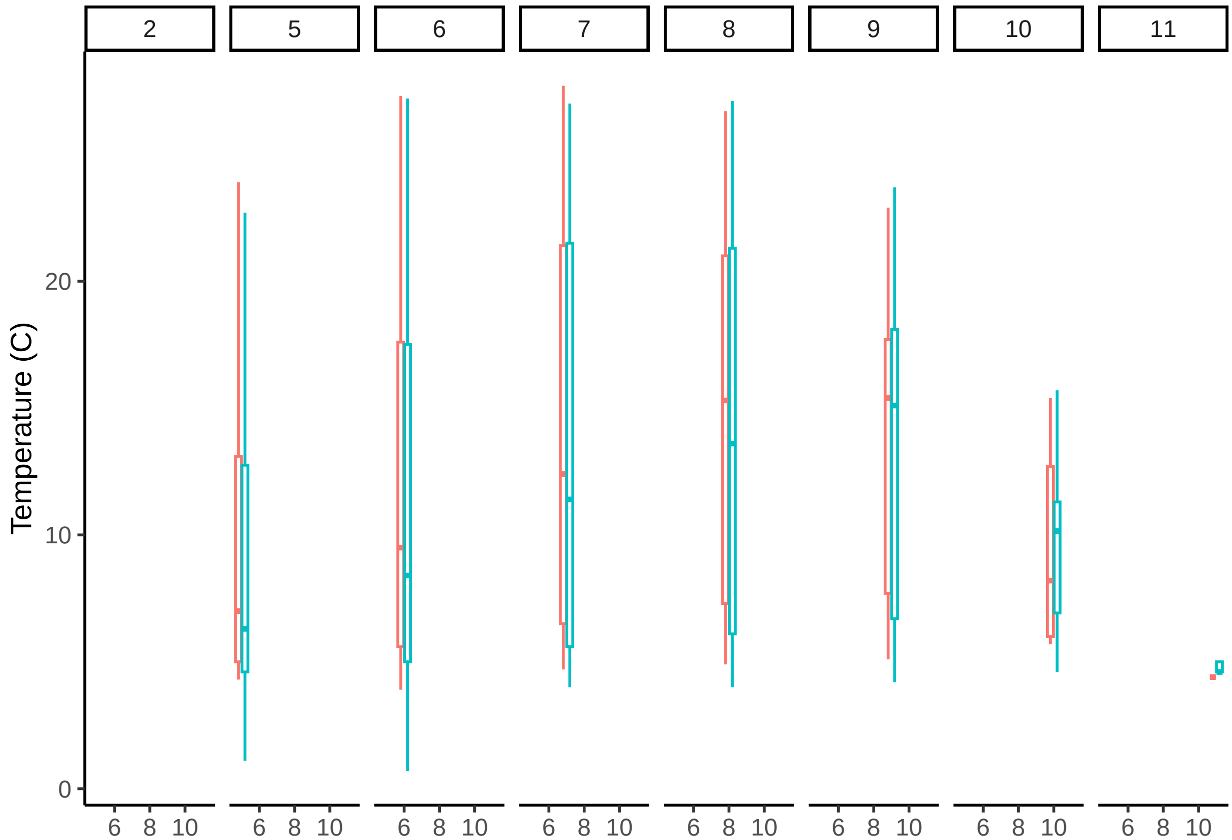


5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned.

Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis title in your theme to `element_blank()` removes the axis title (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different sizes when combined using `cowplot`.

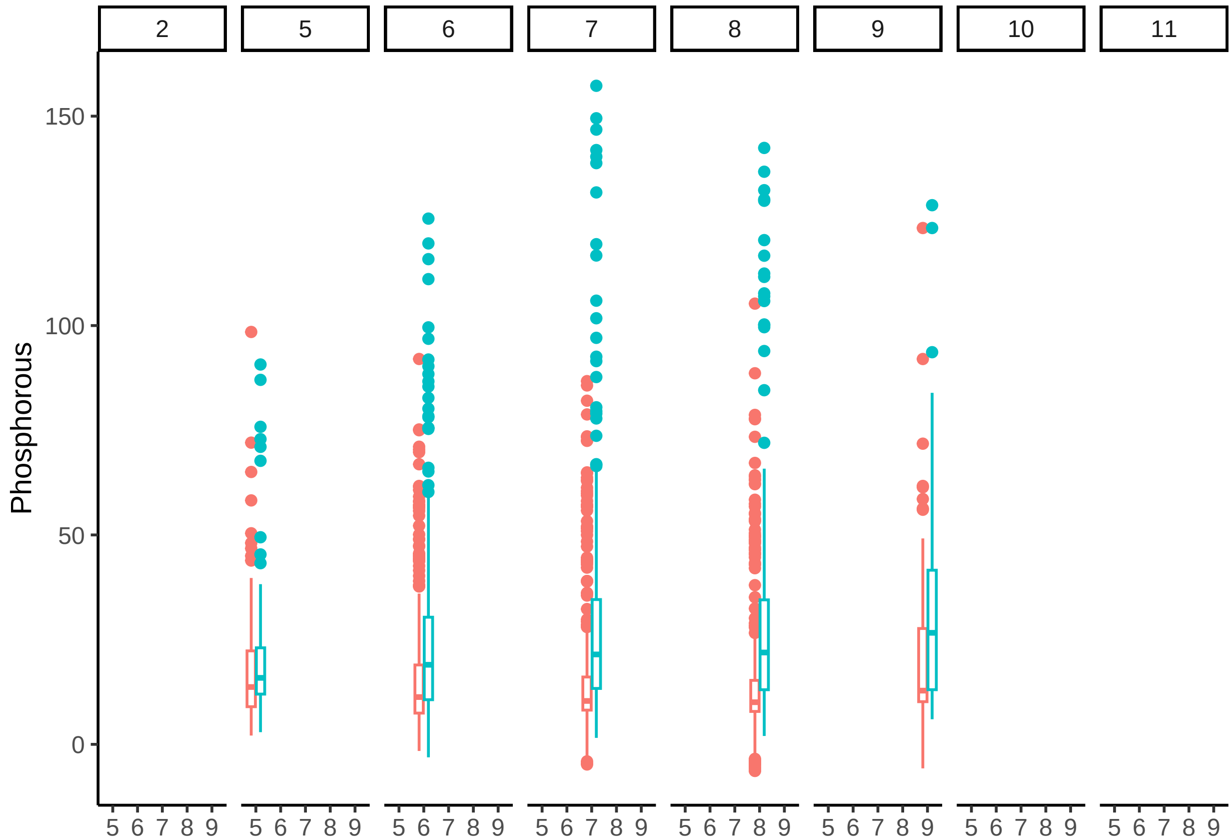
```
#5
temp <-
  ggplot(PeterPaul.chem.nutrients, aes(x = month, y = temperature_C, color = lakename)) +
  geom_boxplot() +
  facet_wrap(vars(month), ncol = 8)+
  labs(x = "Month", y = "Temperature (C)") +
  theme_classic() +
  theme(
    axis.title.x = element_blank(), #remove x axis title
    legend.position = "none" #remove legend
  )
print(temp)
```

```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).
```



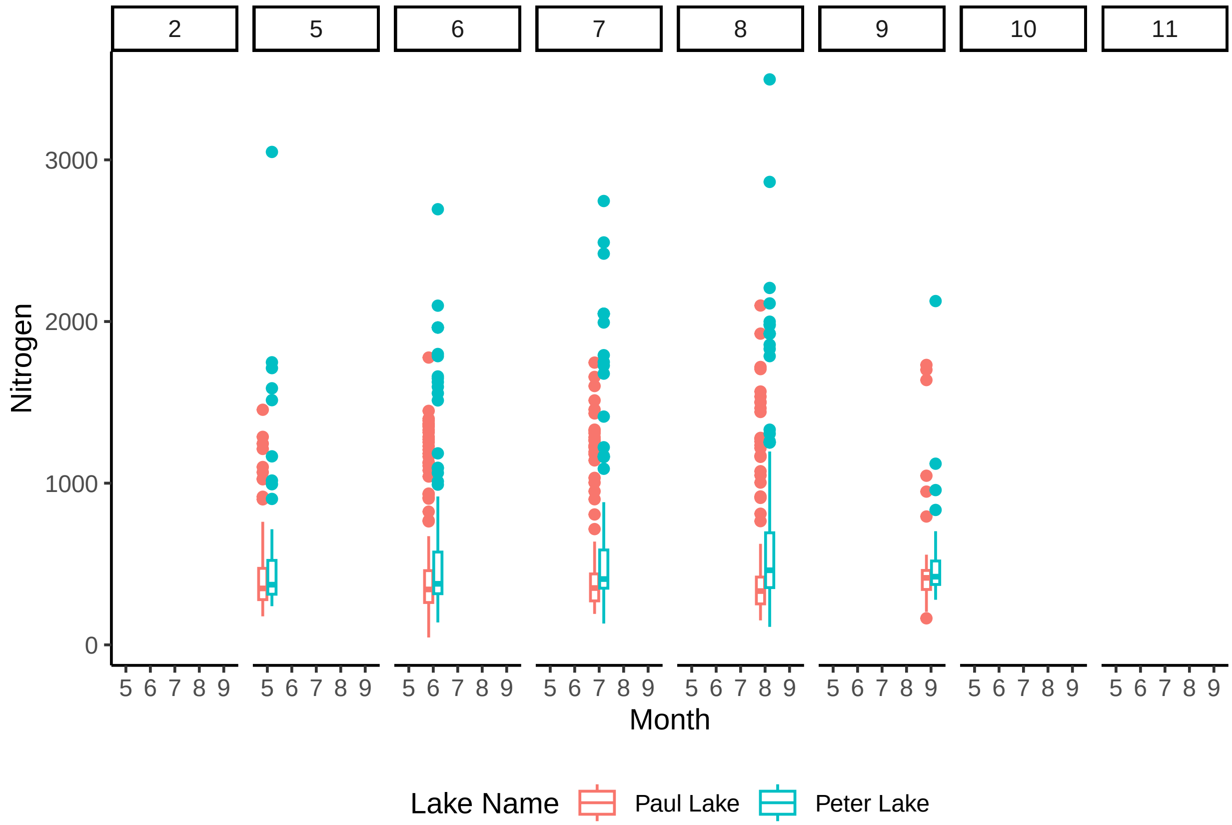

```
tp <-
  ggplot(PeterPaul.chem.nutrients, aes(x = month, y = tp_ug, color = lakename)) +
  geom_boxplot() +
  facet_wrap(vars(month), ncol =8)+
  labs(x = "Month", y = "Phosphorous") +
  theme_classic() +
  theme(
    axis.title.x = element_blank(), #remove x axis title
    legend.position = "none" #remove legend
  )
print(tp)
```

```
## Warning: Removed 20729 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).
```




```r
tn <-
  ggplot(PeterPaul.chem.nutrients, aes(x = month, y = tn_ug, color = lakename)) +
  geom_boxplot() +
  facet_wrap(vars(month), ncol = 8)+
  labs(x = "Month", y = "Nitrogen", color = "Lake Name") +
  theme_classic() +
  theme(legend.position = "bottom")
print(tn)
```

```
## Warning: Removed 21583 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

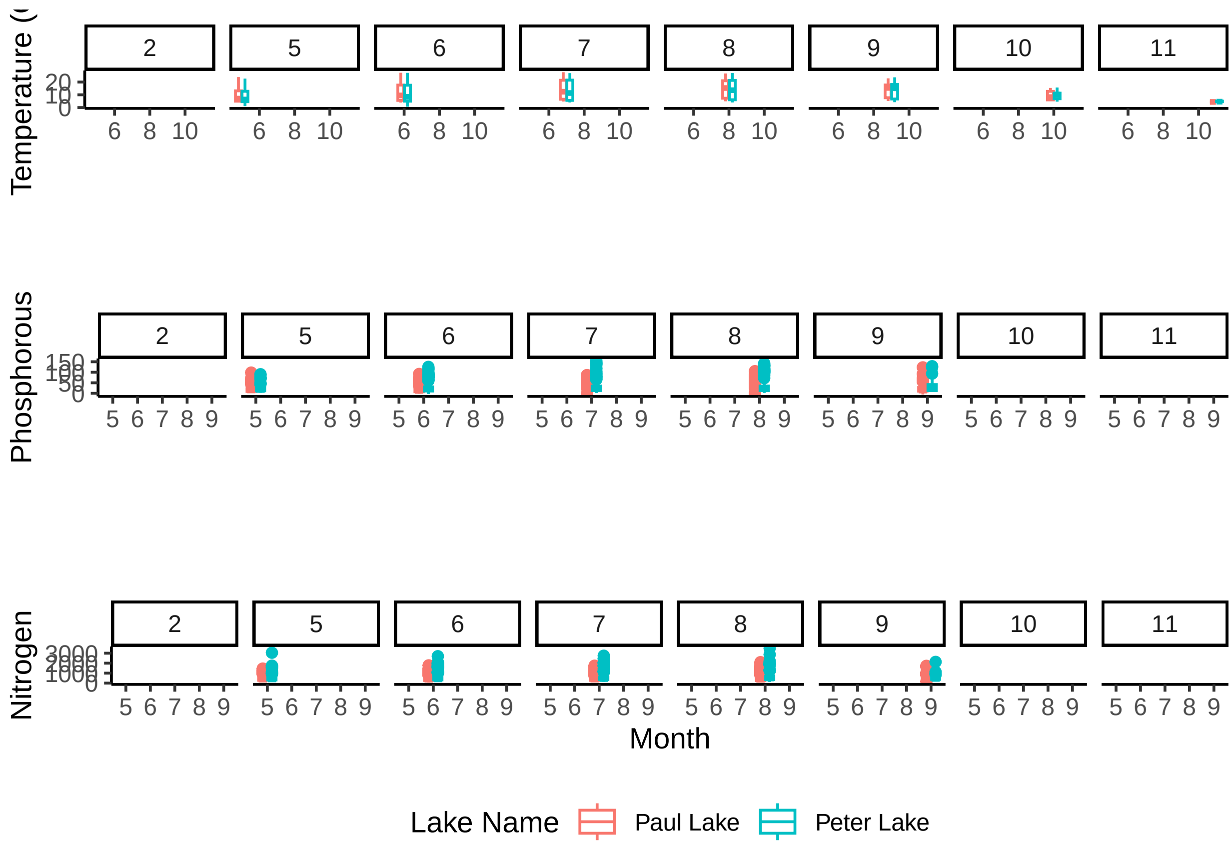



```
plot_grid(temp, tp, tn, nrow = 3, align = 'h')
```

```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).
```

```
## Warning: Removed 20729 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).
```

```
## Warning: Removed 21583 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).
```

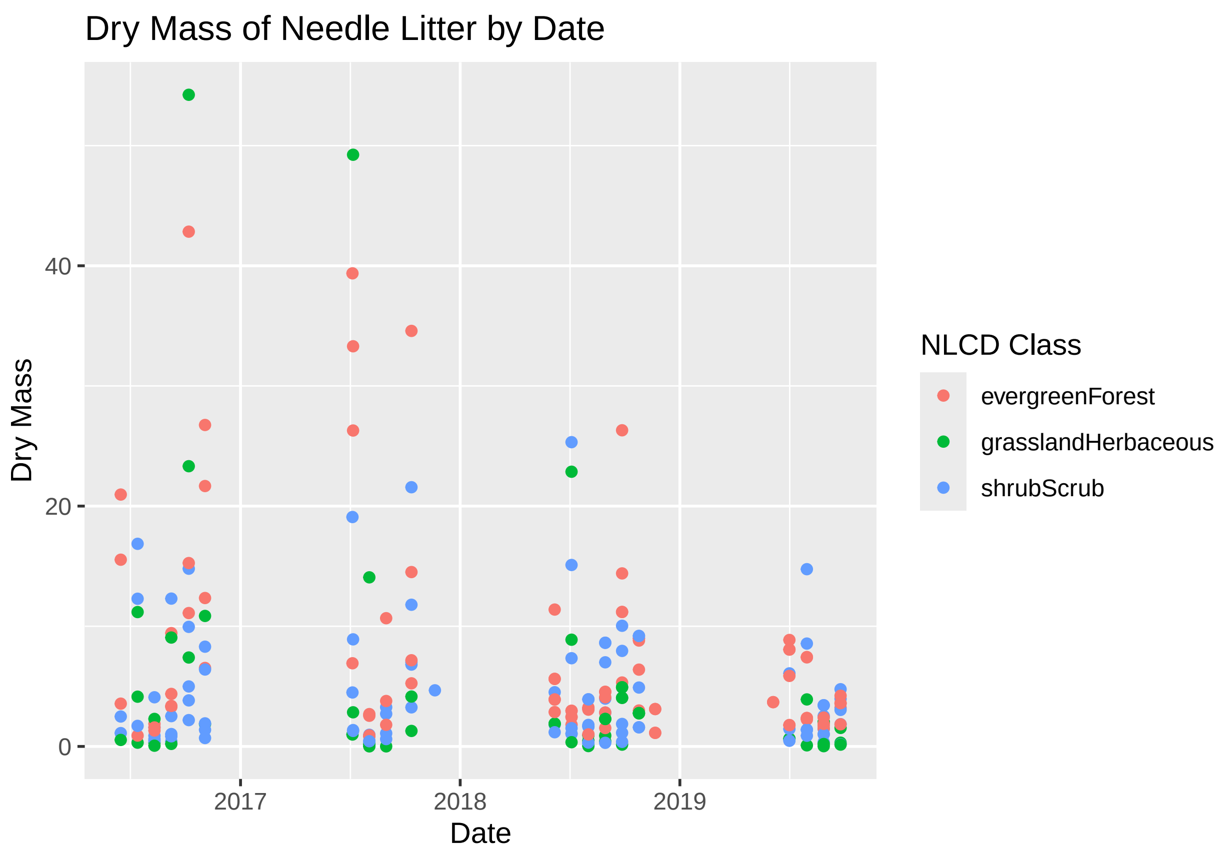


Question: What do you observe about the variables of interest over seasons and between lakes?

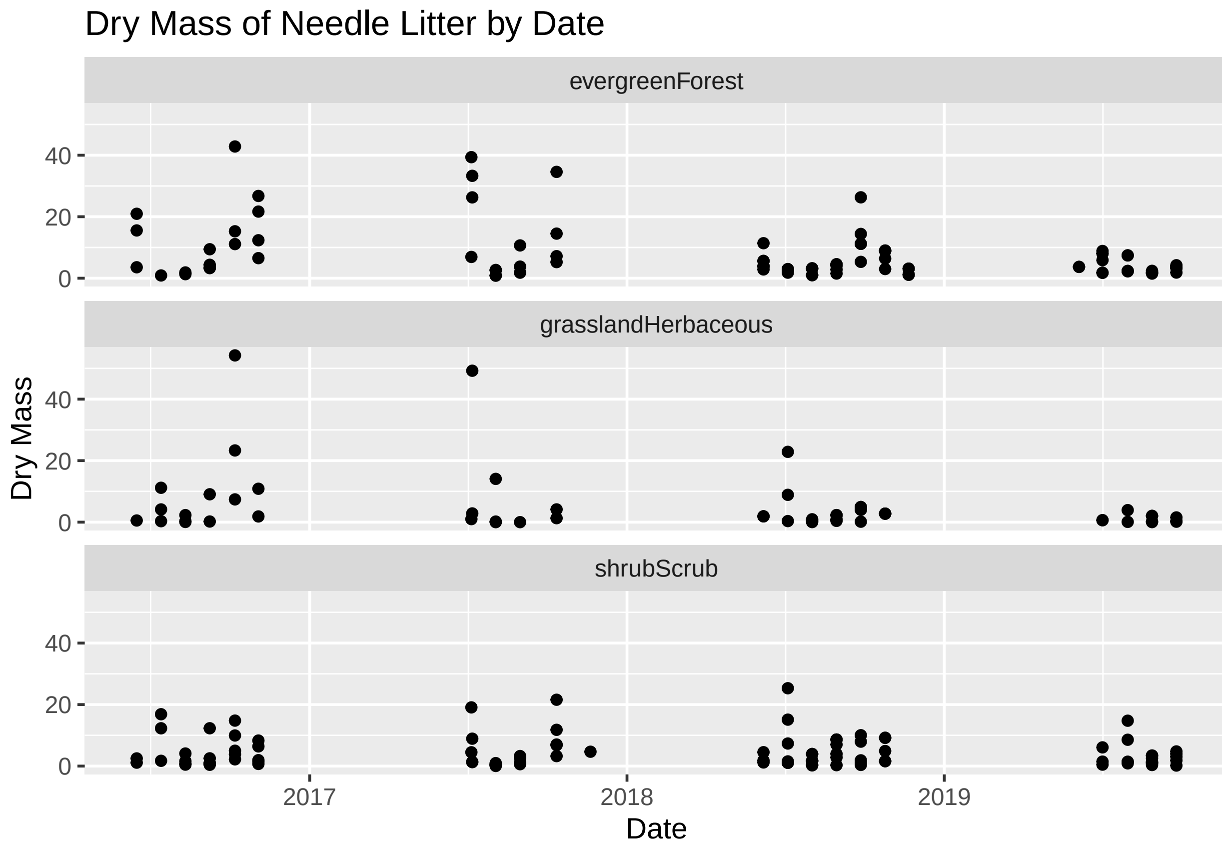
> Answer: Paul Lake has a higher median temperature and lower median levels of phosphorous and nitrogen than in Peter Lake. Facet wrap is used to observe variables across seasons. Lake temperature is warmest in August and September. Phosphorous levels appear relatively contant in Paul Lake, while in Peter Lake levels appear to increase and are are highest in September. Total Nitrogen does not appear to change significantly across seasons.

6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the "Needles" functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)
7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

```
#6 plot dry mass of needle litter by date
drymass_plot <- NIWO.litter.mass %>%
  filter(functionalGroup == "Needles") %>%
  ggplot(aes(x=collectDate,y=dryMass, color=nlcdClass)) +
  labs(title = "Dry Mass of Needle Litter by Date",
       x = "Date", y = "Dry Mass", color = "NLCD Class")+
  geom_point()
print(drymass_plot)
```




```r
#7 facet NLCD
drymass_facet <- NIWO.litter.mass %>%
  filter(functionalGroup == "Needles") %>%
  ggplot(aes(x=collectDate,y=dryMass)) +
  labs(title = "Dry Mass of Needle Litter by Date",
       x = "Date", y = "Dry Mass", color = "NLCD Class") +
  geom_point() +
  facet_wrap(facets=vars(nlcdClass),nrow=3) #facet by NLCD class into 3 rows
print(drymass_facet)
```



Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

> Answer:While plot 7 can be helpful because there is not as much overlap of points, I think plot 6 is more effective because it allows all points to be plotted together using color for an easier comparison.